



LESSON - 9



Operators - Filtering Data

SQL operators are used to perform operations on data in SQL queries. They can be categorized into different types:

ARITHMETIC OPERATORS

Used to perform mathematical calculations on numeric data.

Operator	Description	Example
+	Addition	SELECT 10 + 5; → 15
-	Subtraction	SELECT 10 - 5; → 5
*	Multiplication	SELECT 10 * 5; → 50
/	Division	SELECT 10 / 5; → 2
%	Modulus (Remainder)	SELECT 10 % 3; → 1

COMPARISON OPERATORS

Used to compare values in queries.

Operator	Description	Example
=	Equal to	SELECT * FROM employees WHERE salary = 50000;
!= or <>	Not equal to	SELECT * FROM employees WHERE salary <> 50000;
>	Greater than	SELECT * FROM employees WHERE age > 30;
<	Less than	SELECT * FROM employees WHERE age < 30;
>=	Greater than or equal to	SELECT * FROM employees WHERE salary >= 60000;
<=	Less than or equal to	SELECT * FROM employees WHERE salary <= 60000;
BETWEEN	Between two values (inclusive)	SELECT * FROM employees WHERE salary BETWEEN 40000 AND 80000;
IN	Matches any value in a list	SELECT * FROM employees WHERE department IN ('HR', 'IT', 'Finance');
LIKE	Pattern matching using wildcards (% , _)	SELECT * FROM employees WHERE name LIKE 'J%';
IS NULL	Checks if a value is NULL	SELECT * FROM employees WHERE manager_id IS NULL;
IS NOT NULL	Checks if a value is NOT NULL	SELECT * FROM employees WHERE manager_id IS NOT NULL;

LOGICAL OPERATORS

Used to filter data based on conditions.

Operator	Description	Example
AND	Returns true if both conditions are true	SELECT * FROM employees WHERE department = 'IT' AND salary > 50000;
OR	Returns true if any one condition is true	SELECT * FROM employees WHERE department = 'HR' OR department = 'Finance';
NOT	Negates a condition	SELECT * FROM employees WHERE NOT department = 'Sales';

BITWISE OPERATORS

Used to perform bitwise operations (mostly in databases like MySQL, SQL Server).

Operator	Description	Example
&	Bitwise AND	SELECT 5 & 3; → 1
`	`	Bitwise OR
^	Bitwise XOR	SELECT 5 ^ 3; → 6
~	Bitwise NOT	SELECT ~5; → -6

ASSIGNMENT OPERATORS

Used to assign values.

Operator	Description	Example
=	Assigns a value	SET @total_salary = 50000;

STRING OPERATORS

Used for string manipulation.

Operator	Description	Example
`		or CONCAT() `
LIKE	Pattern matching	SELECT * FROM employees WHERE name LIKE 'J%';
ILIKE	Case-insensitive LIKE (PostgreSQL)	SELECT * FROM employees WHERE name ILIKE 'john%';